

Bolting sugar beet; Source: Umeå
Plant Science Centre/Pierre Pin: *Life
Cycle and Flowering Time Control in Beet*

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Abstract

There are four main genes that were described to regulate the flowering time in beets (*Beta* sp.): *BTC1*, *BBX19*, *FT1*, and *FT2*. We used Illumina whole-genome sequencing data of a large number of sugar beet and wild beet accessions from different geographic regions and analysed the sequence variation at these four loci. We compared the haplotypes present in our data to haplotypes found in previous studies and present a geographic distribution of haplotypes.

Introduction

Just recently over 600 *Beta* accessions have been sequenced, including sugar beet and wild beet species [1]. The data were mapped against RefBeet-1.2 [2] followed by variant calling [3]. This extensive genomic variation data set has been used in this study to analyze haplotype divergence of four important flowering time regulator genes [4]:

- *BTC1 (Bv2_045920_gycn)*
- *FT1 (Bv9_214250_miuf)*
- *FT2 (Bv4_074700_eewx)*
- *BBX19 (Bv9_216430_rwmw/Bv9_216440_udaa)*

The gene identifiers in the reference gene set BeetSet-2 [5] are indicated in brackets.

Single nucleotide variant (SNV) positions within the coding sequence (CDS) of these gene loci were analyzed resulting in allele frequencies at SNV positions, quantification of different haplotypes per gene and per species, and the geographic distribution of haplotypes.

Table 1: Number of sequenced *Beta* accessions per species. Most wild beet accessions have geographic coordinates associated with them.

	Sequenced accessions	Accessions w/ coordinates
<i>Beta vulgaris</i> ssp. <i>vulgaris</i>	318	28
<i>Beta vulgaris</i> ssp. <i>maritima</i>	267	249
<i>Beta vulgaris</i> ssp. <i>adanensis</i>	47	46
<i>Beta macrocarpa</i>	30	25
<i>Beta patula</i>	3	3
Sum	665	351

SNV allele frequencies

To get an overview of the allelic variation of all beet accessions combined, allelic depth from the variant data was used to calculate allele frequencies of the SNVs for each of the four genes. SNVs with frequencies below 2 % were removed to reduce the number of false positives due to sequencing errors. Positions were numbered relative to the translation start site, along with the number of the corresponding exon. The allele frequencies of gene *FT1* are shown as example (Figure 1).

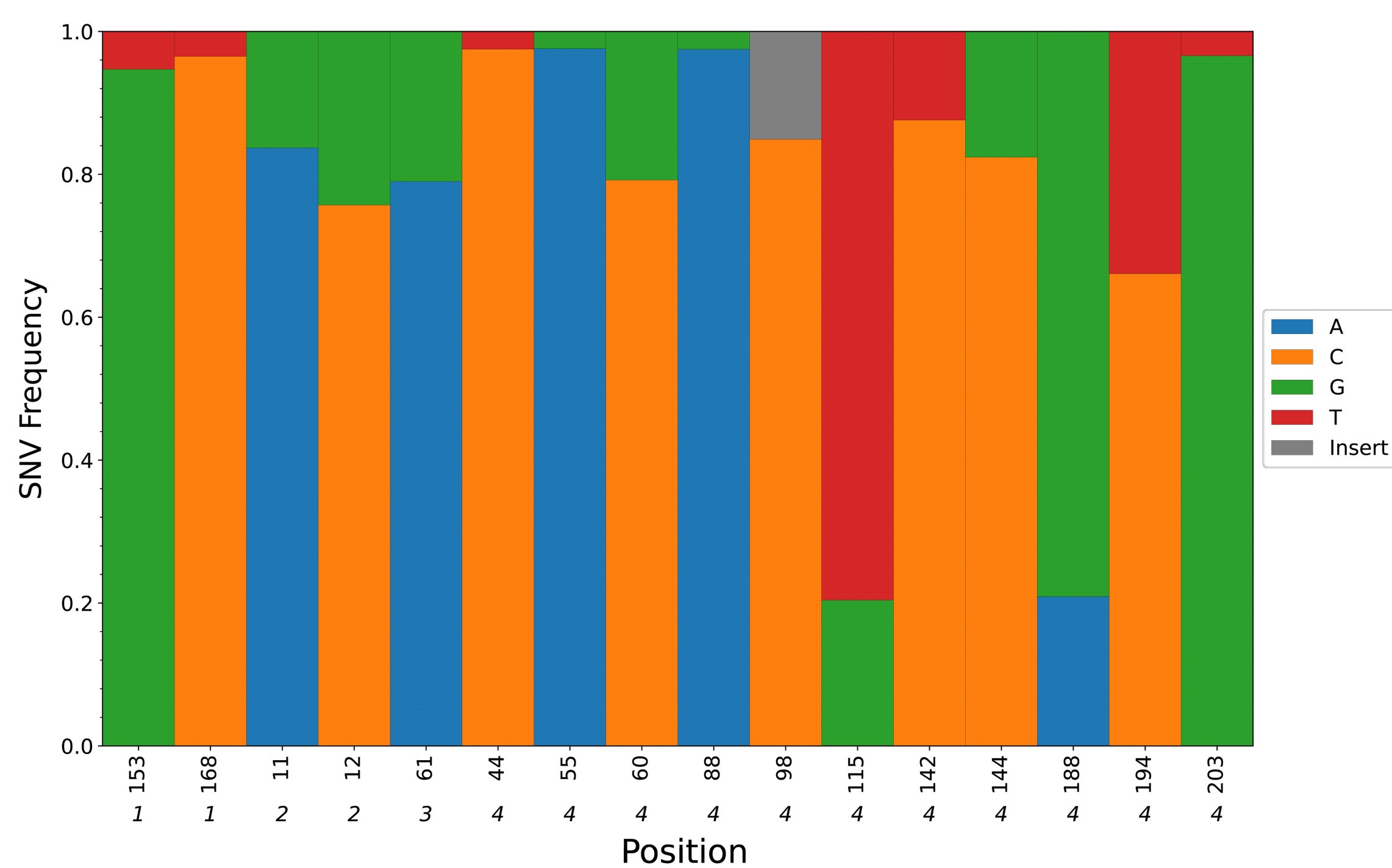


Figure 1: Allele frequencies at SNV positions in gene FT1 combined for all *Beta* accessions analyzed. Vertical numbers in the bottom line indicate the exon number. Horizontal numbers on top specify the position within each exon.

Allele frequencies in sugar beet and wild beet

In addition to the stacked barplots for the complete data, we also determined allele frequencies for each species separately. The example below facilitates the visual comparison between sugar beet and sea beet. It is noteworthy that the allele frequencies in cultivated beets (*B. v. vulgaris*) were quite similar in sea beets (*B. v. maritima*), differing only in a few places (Figure 2).

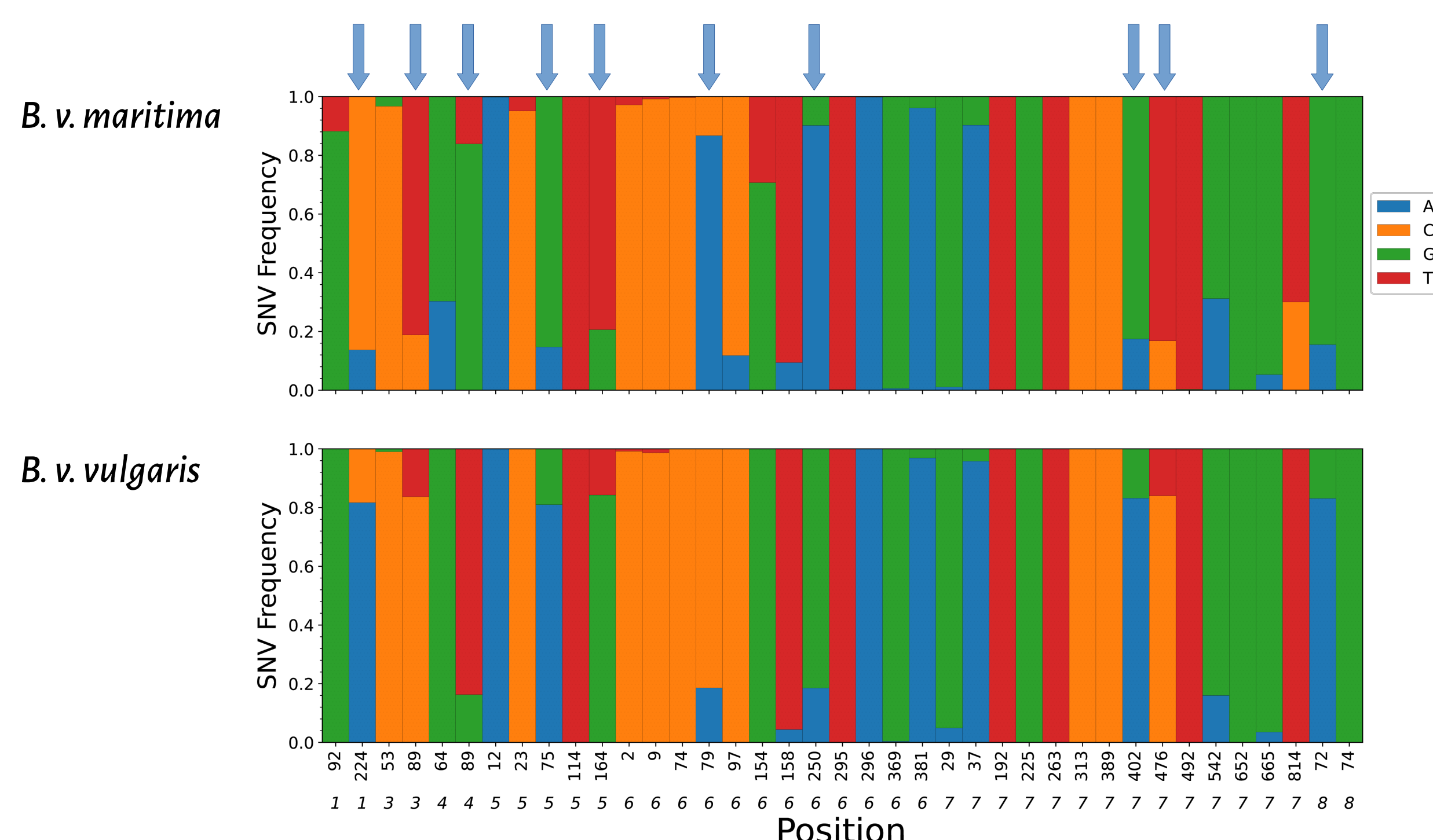


Figure 2: Allele frequencies in gene *BTC1*, separately for *B. v. maritima* (top) and *B. v. vulgaris* (bottom). SNV positions are given relative to the translation start site, with the number of the corresponding exon below. Positions showing differing allele frequencies between the two subspecies are indicated by arrows.

Number of different haplotypes

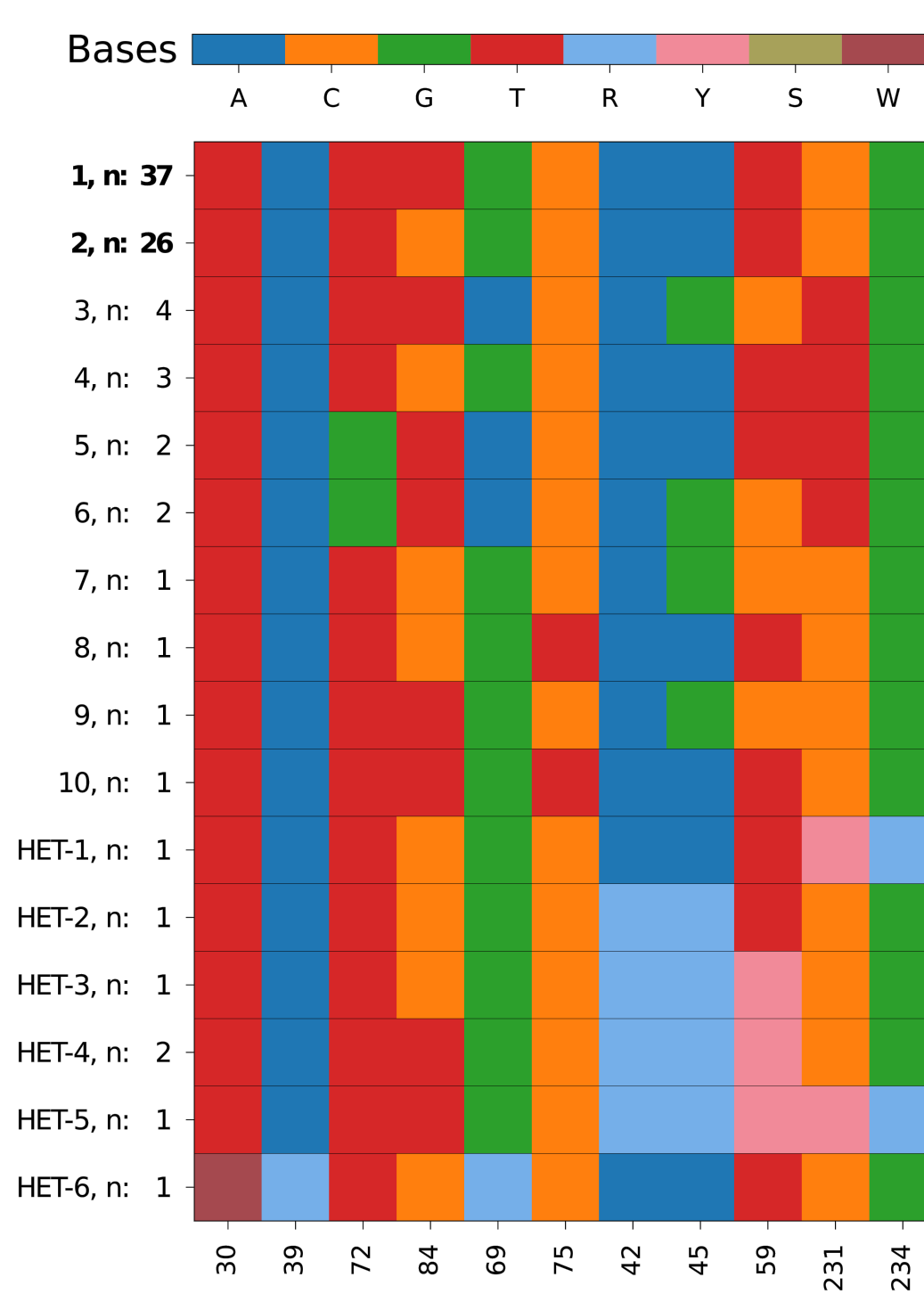


Figure 3: Haplotype matrix for *BBX19* in *B. v. adanensis*. The value after 'n' indicates the number of occurrence of this haplotype (assuming diploidy).

To identify and quantify the most common haplotypes in each species, accessions were grouped together by identical alleles across variant positions per gene. If there was a single heterozygous position, those ‘ambiguous’ accessions were resolved by splitting them into two haplotypes. As a consequence, accessions without heterozygous positions were counted twice, assuming a diploid genome.

The result for *BBX19* in *B. v. adanensis* is shown in Figure 3. Haplotypes were sorted by count, and remaining ambiguous haplotypes, i. e. those with more than one heterozygous position, were appended to the matrix. The extended IUPAC nucleotide nomenclature was used.

Compared to Hóft et al. (2018), more SNVs and haplotypes were found for each gene in our extended data set. This is to be expected due to the larger number of accessions (665 vs. 29) and a wider range of wild beet species analyzed (*B. v. adanensis*, *B. macrocarpa* and *B. patula* in addition to *B. v. maritima*). Some of SNVs determined previously were not present in our data (Table 2). It is worth noting that only a small number of different haplotypes were confirmed by a high count. Rare haplotypes inferred from our data may be artifacts due to sequencing errors.

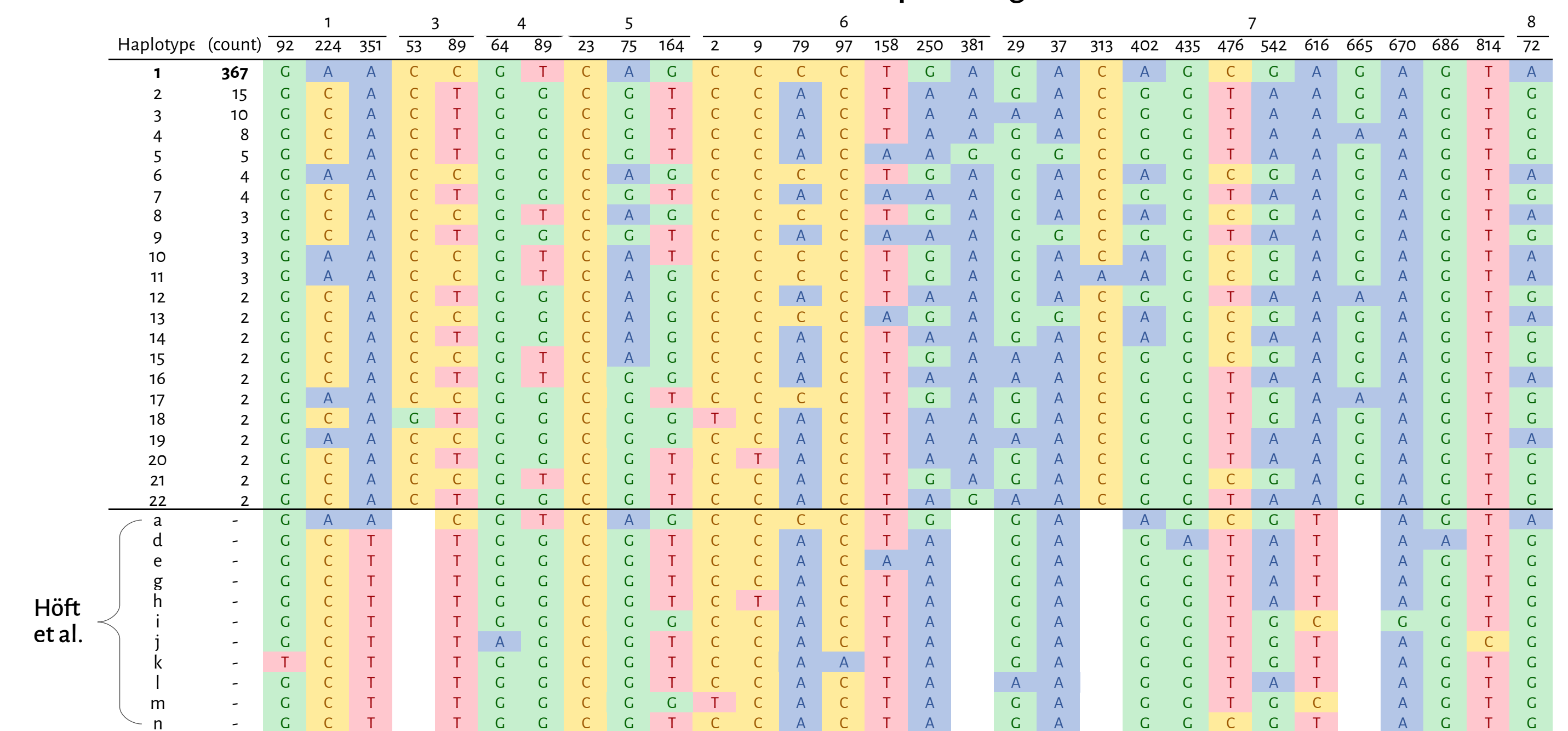
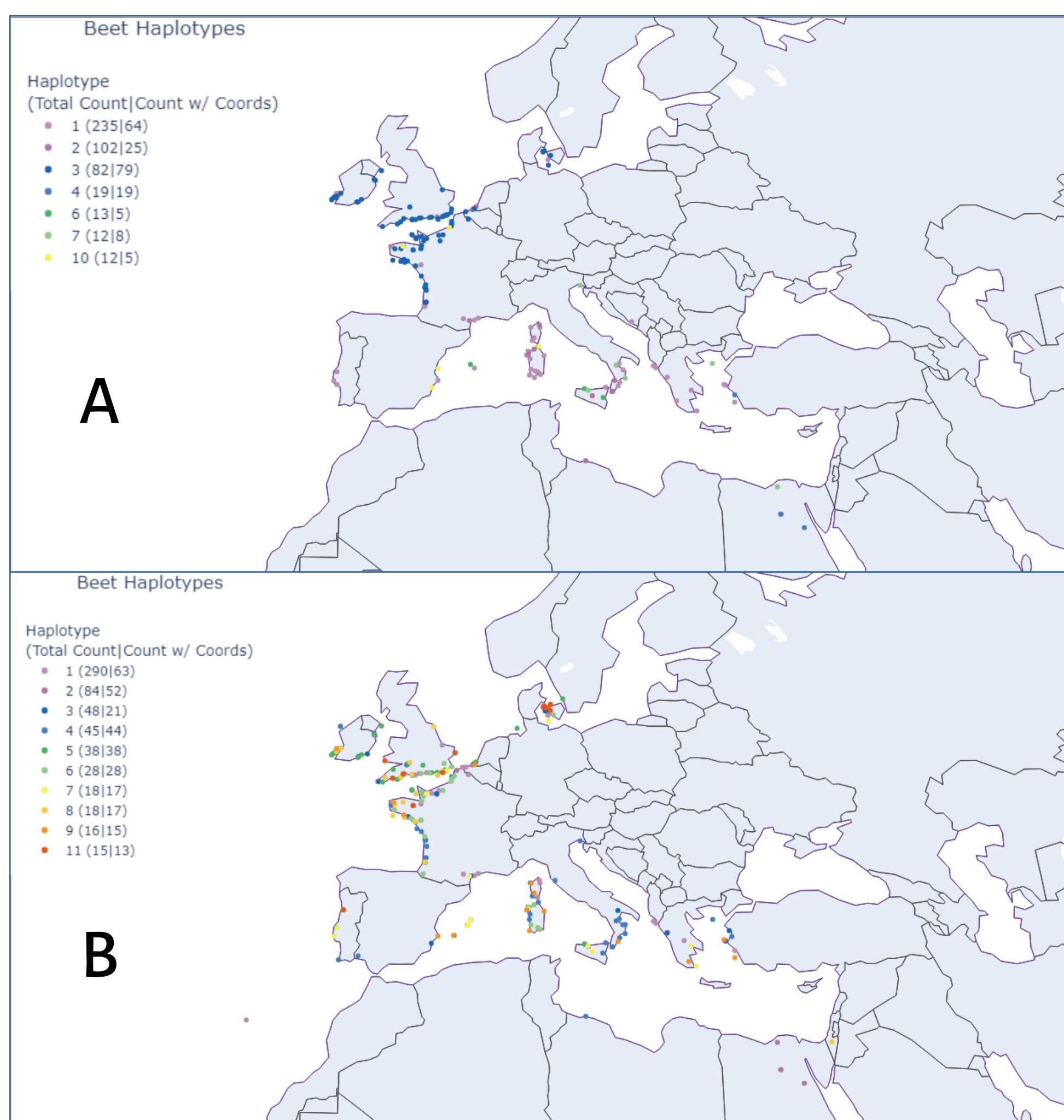


Table 2: Summary of haplotypes for *BTC1* in *B. v. vulgaris*. Haplotypes identified in this study (at least two occurrences) are shown on top, haplotypes found by Höft et al. (2018) are shown at the bottom.

Geographic distribution of haplotypes



Since geographic coordinates were associated with many of the accessions analyzed we developed an interactive tool for haplotype display on a geographic map. The tool allows for selection of the gene, filtering by a minimum count of accessions in a haplotype, and subsetting by species.

In general, the haplotypes were clustered in certain regions, e.g., the Mediterranean, the southern Iberian Peninsula, or the northern French-Great Britain coast (Figure 4 A). However, some haplotypes were distributed throughout Europe (Figure 4 B).

Figure 4: Geoplots of the haplotypes for genes *FT1* (A) and *FT2* (B) in *B. v. maritima*. The filter for minimum count of accessions in a haplotype was set to 12. Haplotypes were ordered by total count of accessions.

Summary and conclusion

- SNVs in four important flowering time regulator genes of beets feature wide variation across a broad range of *Beta* accessions, including sugar beets and wild beet species.
- Accessions were grouped by haplotype, and major haplotypes can be identified by number of occurrences.
- By creating an interactive map of haplotypes, we show that haplotypes are generally grouped into regions, with some haplotypes distributed throughout Europe.

References

- [1] Wascher et al. (2022). Genomic distances reveal relationships of wild and cultivated beets. *Nat Commun.* 2022 Apr 19;13(1):2021.
- [2] Dohm et al. (2014). The genome of the recently domesticated crop plant sugar beet (*Beta vulgaris*). *Nature.* Jan 23;505(7484):546-9.
- [3] Felkel et al., submitted
- [4] Höft et al. (2018). Haplotype Variation of Flowering Time Genes of Sugar Beet and Its Wild Relatives and the Impact on Life Cycle Regimes. *Frontiers in plant science*, 8, 2211;
- [5] Minoche et al. (2015). Exploiting single-molecule transcript sequencing for eukaryotic gene prediction. *Genome Biol.* 2015 Sep 2;16(1):184.